

GENITAL SYSTEM

MALE GENITAL SYSTEM

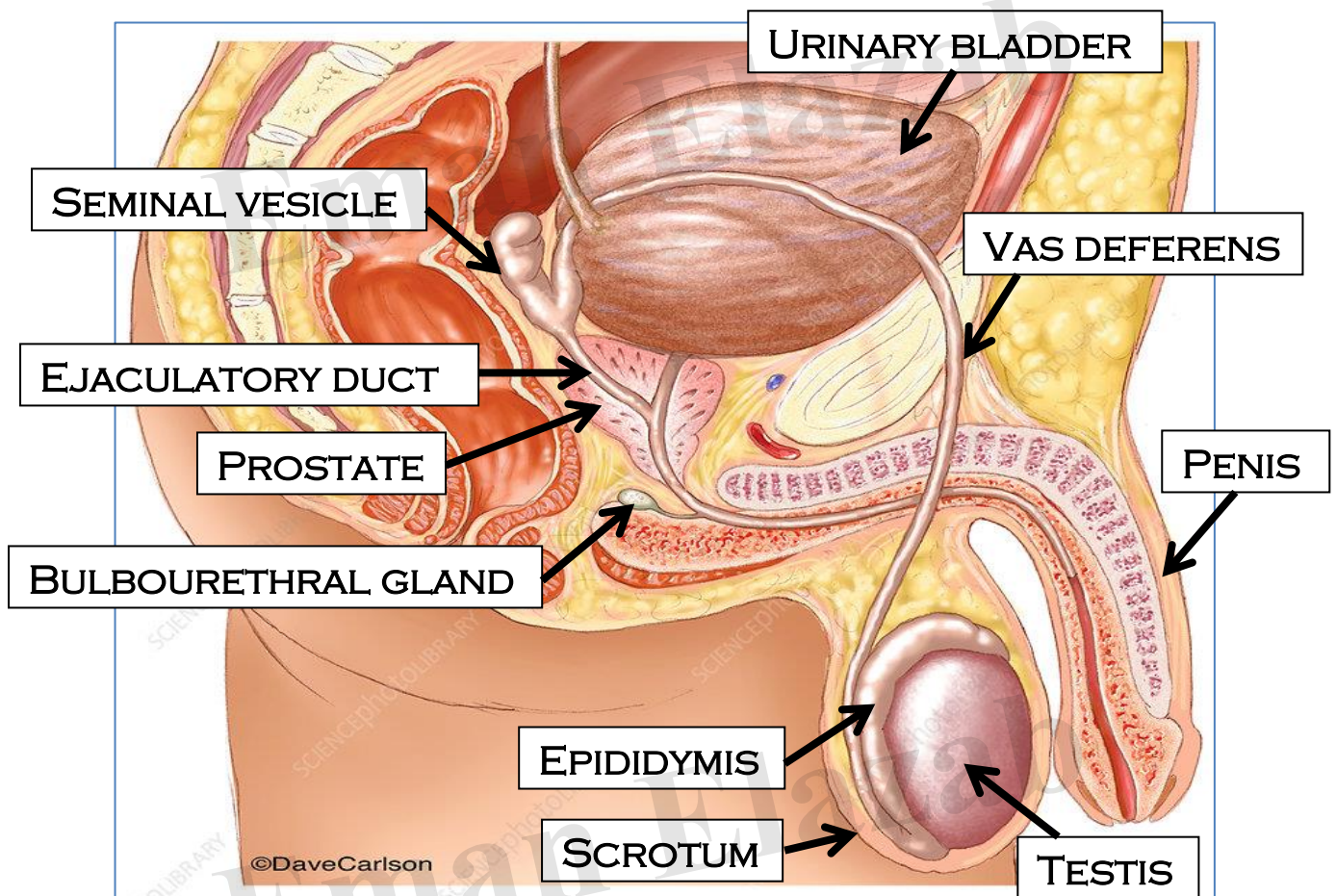
★ It consists of:

A- Testis.

B- Genital ducts.

C- Accessory sex glands.

D- External genitalia: include penis and scrotum.



Testis

- It is the **primary sex organ**.
- It is oval in shape and present inside skin sac called **scrotum**.
- It has 2 poles (upper and lower), 2 surface (medial and lateral) and 2 borders (anterior and posterior).

- **Function: It is a mixed gland.**

- 1-Formation of the mature sperms.

- 2-Secretion of male sex hormone (testosterone hormone).

Genital Ducts

1- Epididymis

- An extremely coiled tubule forming **coma-shaped** structure composed of **head, body and tail**.

- It is **related to** the posterior border of the testis.

- **Function**

- a- **Storage and maturation** of the sperms.

2- Vas deferens

- A **thick-walled narrow** duct, about **45 cm** in length.

- It is the **continuation of** the tail of epididymis, enters the **pelvis** where it **unites** with the seminal vesicle behind the urinary bladder to form the **ejaculatory duct**.

- **Function:** It **transports** the sperms.

3- **Ejaculatory duct:** it opens in the prostatic urethra.

4- **Urethra:** It **transports** the sperms to outside.

Accessory Sex Glands

1- Seminal vesicle

- A highly tortuous tube, lies on the **back (base) of the urinary bladder**.

- It **unites with vas** deferens to form the **ejaculatory duct** which pierces the **prostate** to opens in the **prostatic** part of the **urethra**.

- **Function:** It **secretes nutritive substances** rich in protein, fructose and vitamin C to help motility of sperms.

2- Prostate

- It is a **pyramidal** shaped gland, present inside the pelvic cavity and surrounds the proximal 3 cm of male urethra.
- Its **base** is directed **upwards**, surrounding the **neck urinary bladder**.
- It **opens** by many ducts on the prostatic urethra.
- **Function:** it secretes **alkaline secretion** to neutralize the acidity of vagina.

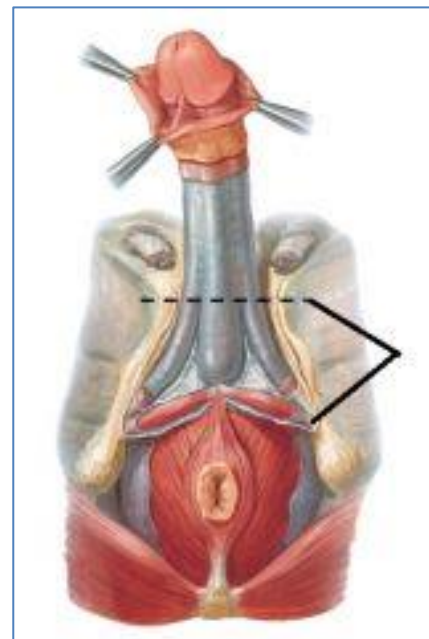
3- Bulbourethral gland

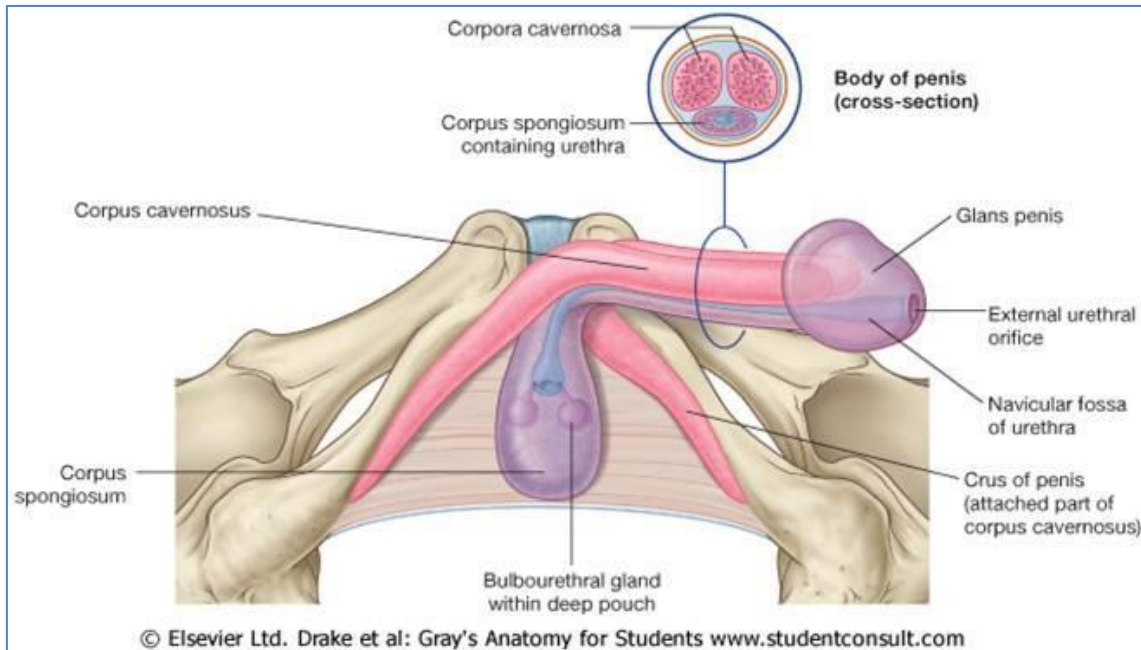
- A small gland lies beside the **membranous urethra**.
- It secretes **mucous to lubricate** the urethra.

★ **N.B.: Semen consists** of sperms and secretions of epididymis ,seminal vesicle and prostate .

Penis

- It is an **erectile organ** for sexual intercourse.
- It **consists of** 3 elongated erectile masses, 2 dorsally: corpora cavernosa and one ventrally corpus spongiosum which contains the penile urethra.
- The penis has **2 parts**: Root (the hidden part) and body (the free pendulous part).
- The **distal expanded part** of the body is called **glans penis** which is **covered by** skin fold called **prepuce**.





FEMALE GENITAL SYSTEM

★ It consists of:

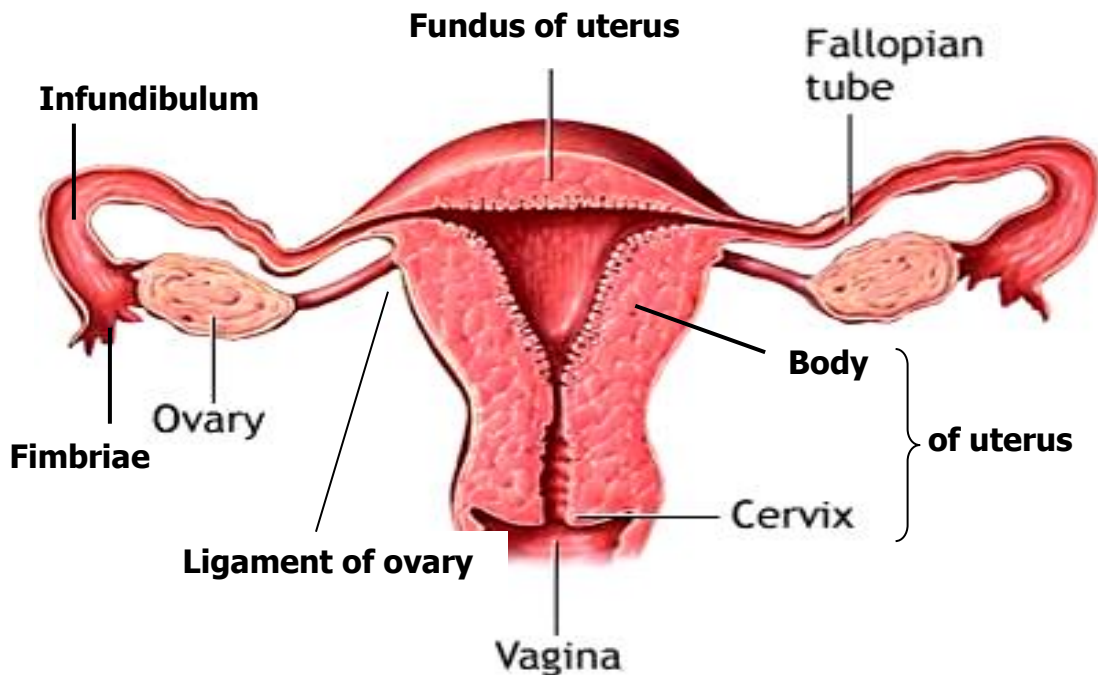
1- Ovary: Oval (almond) shaped, measuring **3 x 2 x 1 cm** attached to the upper lateral angle of the uterus by the **ligament of ovary**.

- a- **Function:** Production of **mature ova** (female germ cells).
- b- Secretion of **female sex hormones:** estrogen and progesterone.

2- Uterine (Fallopian) tubes: **10 cm** tube **opens in** the upper lateral part of uterine cavity **connecting** it with the ovary. Its **lateral end** is called infundibulum (funnel) which has many **fingers-like processes (fimbriae)** extend around the ovary.

- **Function of uterine tube**

- a- It **receives the ovum** from the ovary.
- b- **Passage for the sperms** from uterus to the ovum.
- c- **Fertilization** (the union of one sperm and ovum) occurs in its **lateral 1/3**.
- d- Transport the early embryo to the uterine cavity.



3- Uterus: A **hollow pear-shaped** organ with **thick muscular wall**. It measures **3 x 2 x 1 inches**.

- It is divided into **3 parts**:
 - a- **Fundus:** the part of uterus above the level of uterine tubes.
 - b- **Body:** the part below the level of the uterine tubes.
 - c- **Cervix:**
 - The lower narrow part which projects into the upper part of the vagina.
 - It has supra-vaginal part and intra-vaginal parts
- **Function of uterus**
 - 1) Reception, retention and nutrition of the fertilized ovum until the formation of full term **fetus**.
 - 2) It is the site of the **menstrual cycle**.

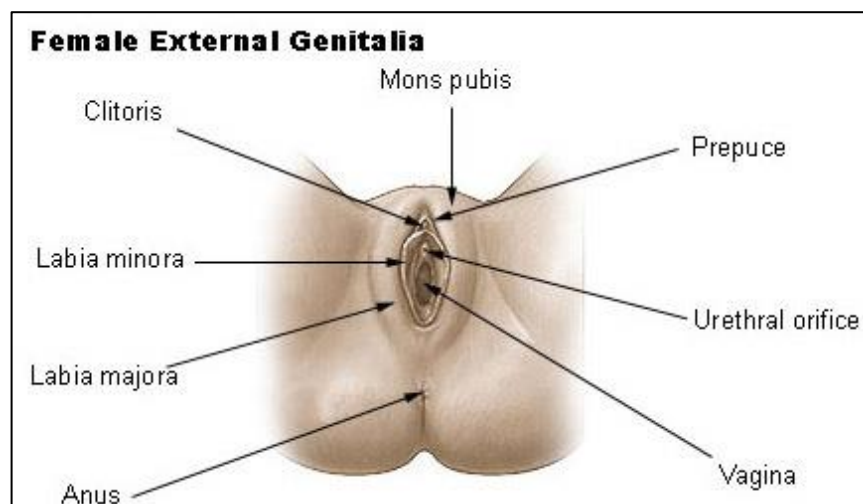
4- Vagina

- **Muscular tube, extends** upwards and backwards **from the vulva** (female external genital organ) **to the uterus**.
- Its anterior wall is short and its **posterior wall is longer**.
- The **upper end** is pierced by the cervix of uterus and the **lower end** is the vaginal orifice.

- **Function of vagina**

- a- **Passage** of menstrual blood.
- b- Birth canal.
- c- Sexual intercourse to introduce the sperms.

4- Female External Genital Organs (Vulva): It consists of:



Mons pubis: An anterior hairy thick skin fold, having a collection of subcutaneous fat.

Labia majora: Two rounded hairy thick skin folds, one on each side, having a collection of subcutaneous fat.

Pudendal cleft: The space between the 2 labia majora.

Labia minora: Two non-hairy thin skin folds, one on each side.

Vestibule of vagina: The space between the 2 labia minora. It contains orifices of urethra and vagina.